

Evaluation Setup & Reporting Metrics

Metric definitions

nRMSE

$$\frac{\text{RMSE}_{\text{well}}}{\text{IQR}_{\text{well}}}$$

Primary metric. Scale-invariant.
Robust to well-level amplitude differences.

NSE

$$1 - \frac{\sum (\hat{y} - y)^2}{\sum (y - \bar{y})^2}$$

Nash-Sutcliffe efficiency.
0 = mean baseline, 1 = perfect fit.

RMSE

$$\sqrt{\text{mean}((\hat{y} - y)^2)}$$

Root mean squared error in original GWS units.

Aggregation strategy

- Per-well: nRMSE and NSE computed for each well independently over time
- nRMSE_pw: median of per-well nRMSE ← headline metric
- Per-horizon: metrics broken down by forecast week $h = 1 \dots 16$
- Clipping: outlier wells capped at nRMSE = 5 before median

Pipeline	nRMSE_pw	nRMSE_pw (clipped)	% clipped
GRU all_train (in-sample ceiling)	0.313	—	—
Oracle GP (true GWS as GP input)	1.666	1.666	0.1%
Decoupled GRU → GP (95 / 5 split, rmd90_test52)	1.975	1.959	5.5%
Joint GRU + GP (λ = 0.001) (90 / 10 split, rmd90)	2.281	2.146	7.7%

★ nRMSE_pw (clipped) = headline metric after capping outlier wells at nRMSE = 5

